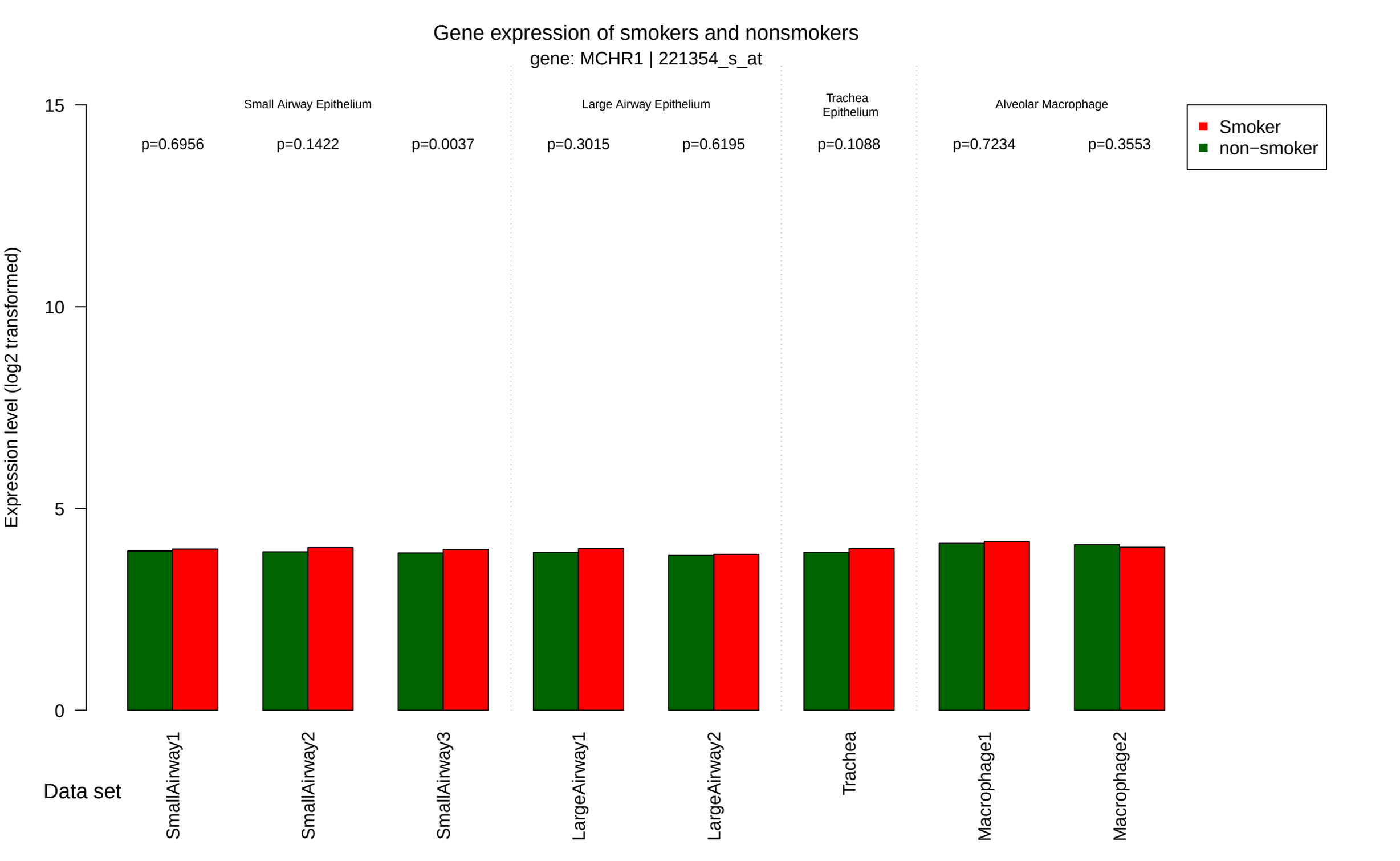
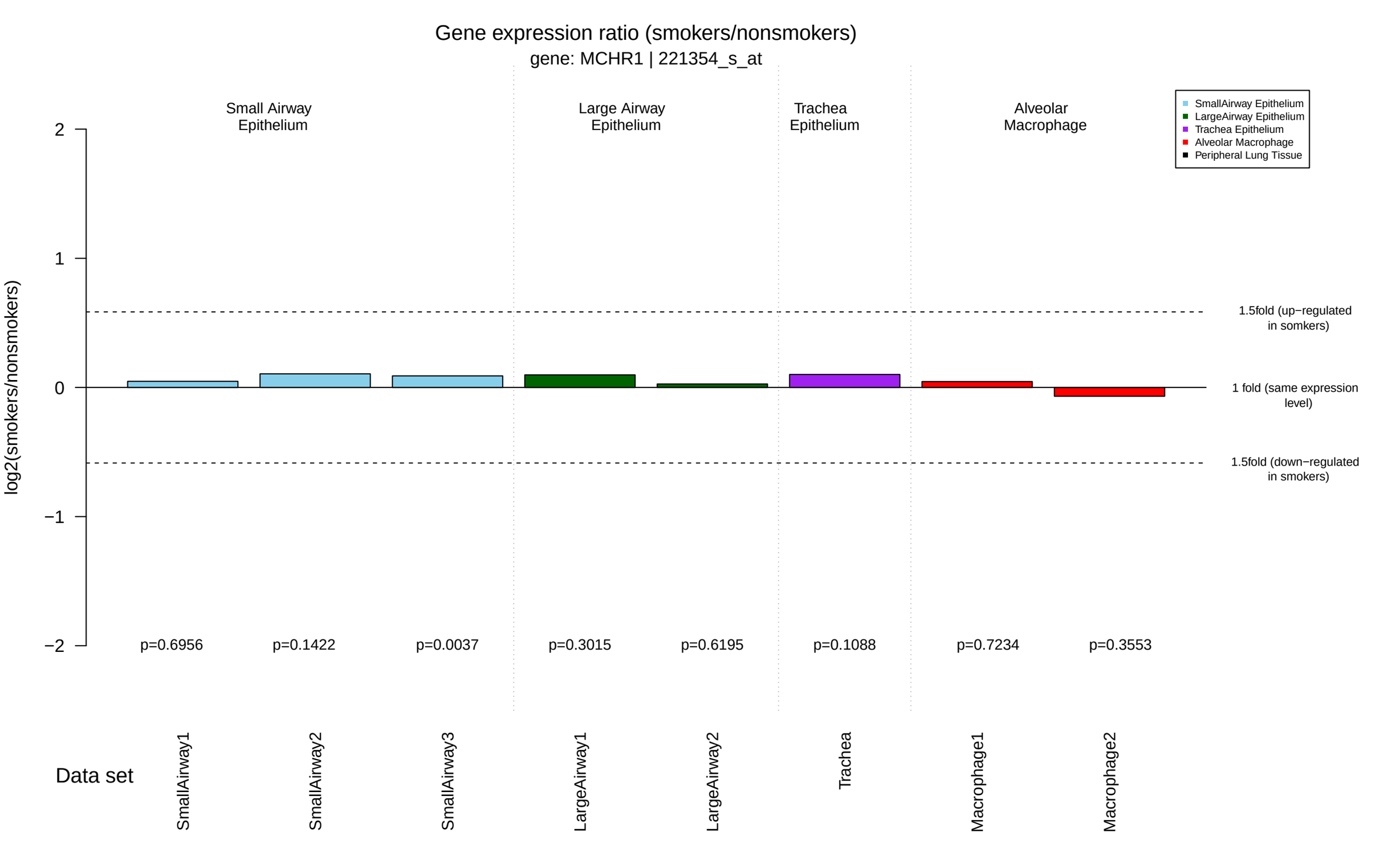
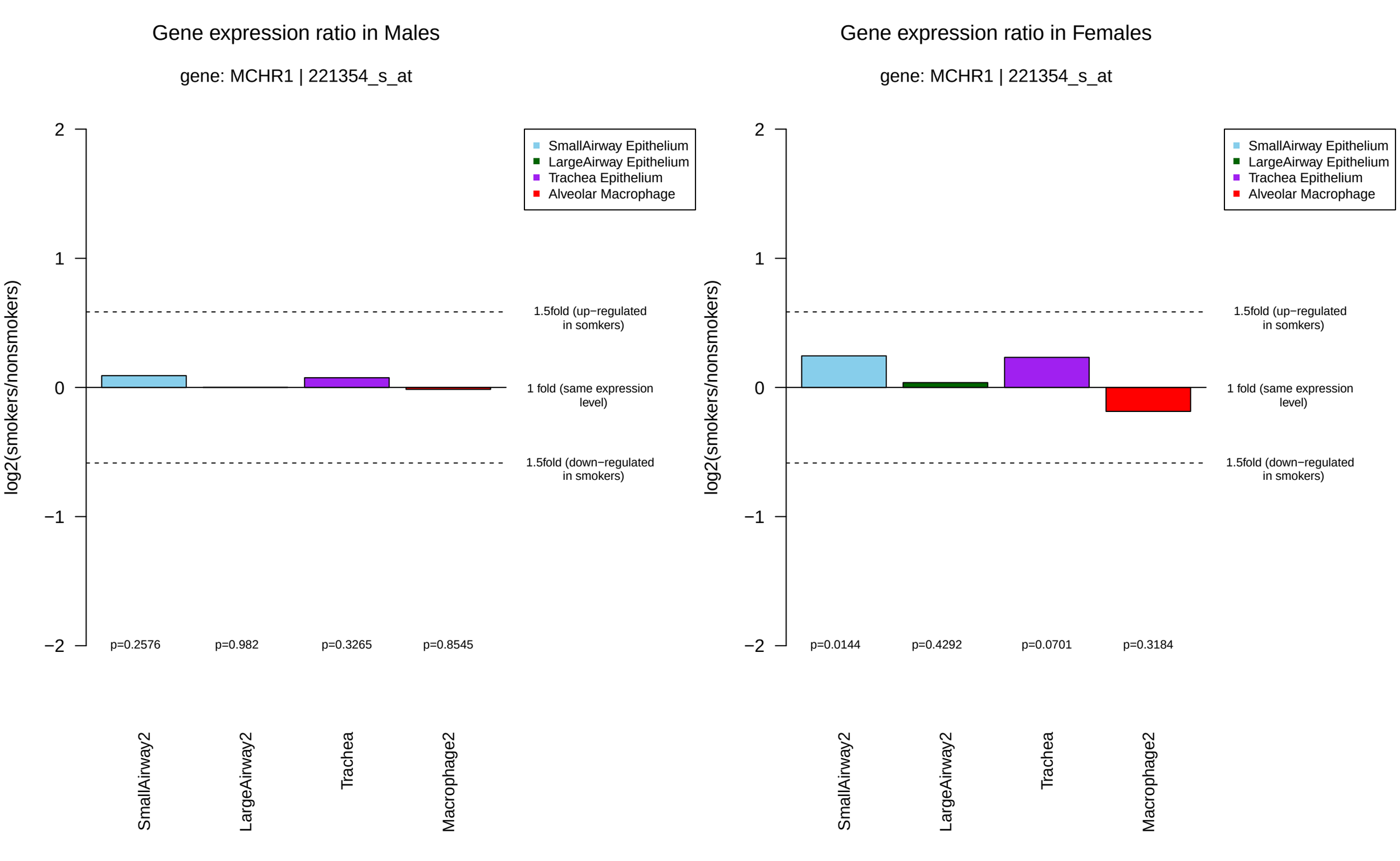
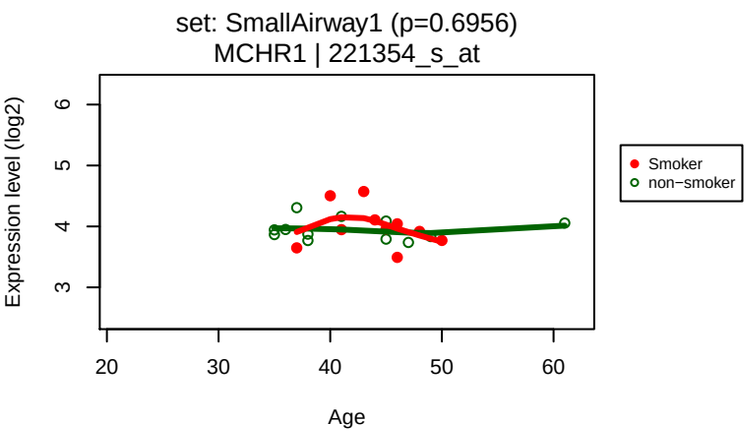
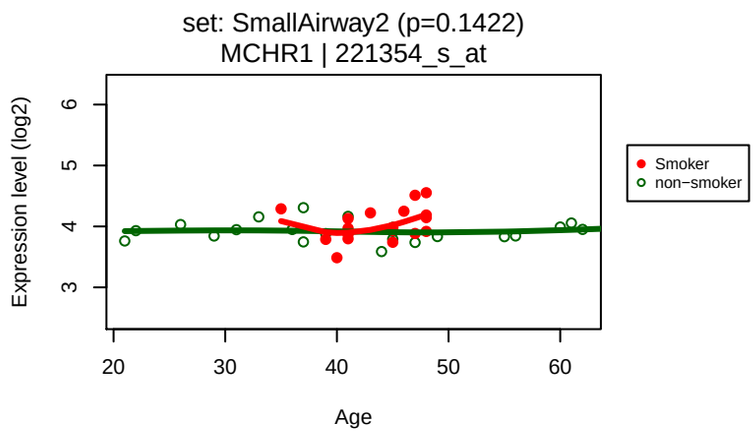
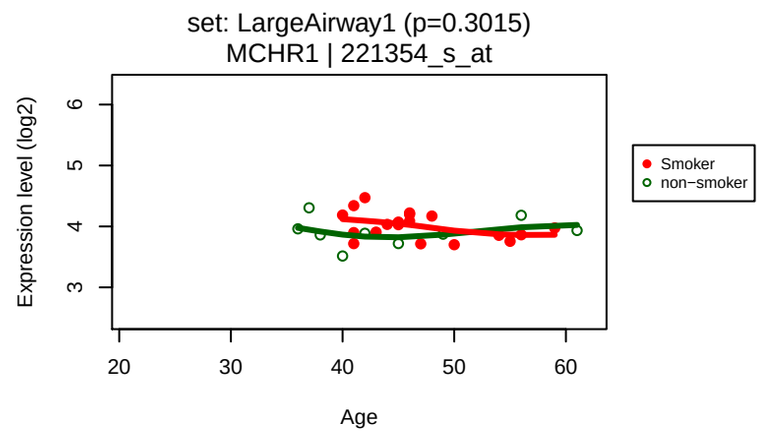

Gene expression of smokers and nonsmokers
gene: MCHR1 | 221354_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6956
p=0.1422
p=0.0037
p=0.3015
p=0.6195
p=0.1088
p=0.7234
p=0.3553
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

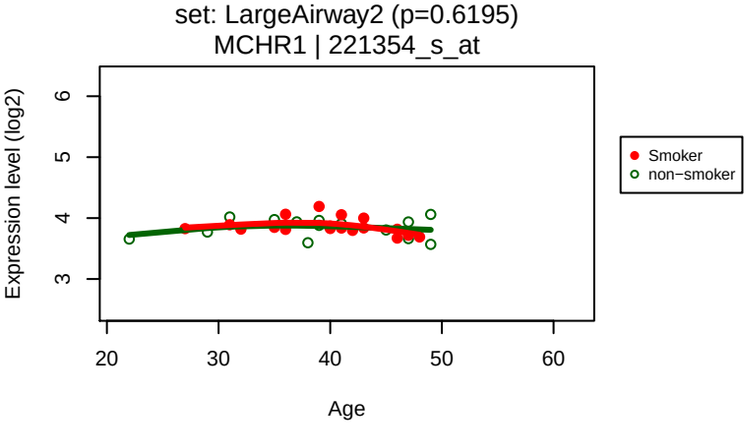

Gene expression ratio (smokers/nonsmokers)
gene: MCHR1 | 221354_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6956
p=0.1422
p=0.0037
p=0.3015
p=0.6195
p=0.1088
p=0.7234
p=0.3553
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

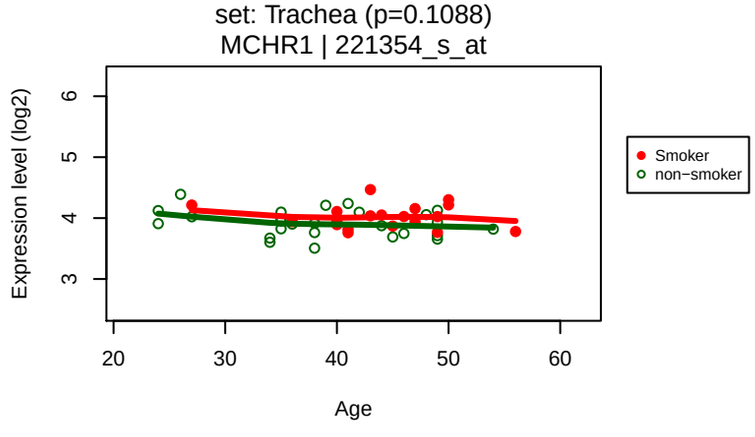

Gene expression ratio in Males
gene: MCHR1 | 221354_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2576
p=0.982
p=0.3265
p=0.8545
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: MCHR1 | 221354_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0144
p=0.4292
p=0.0701
p=0.3184
SmallAirway2
LargeAirway2
Trachea
Macrophage2

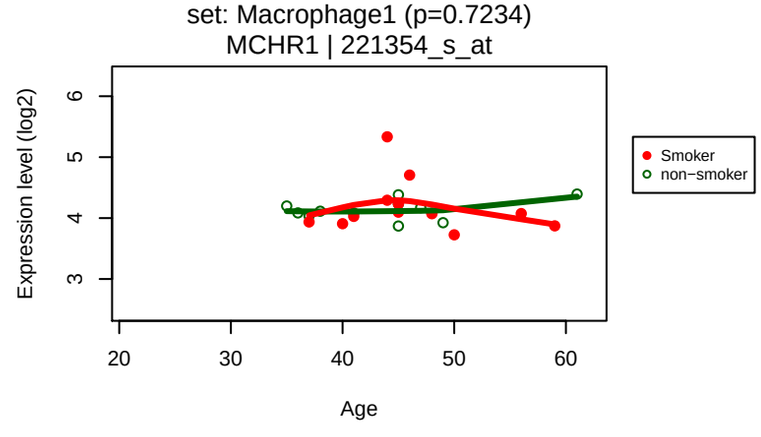

set: SmallAirway1 (p=0.6956)
MCHR1 | 221354_s_at
Expression level (log2)
Age
Smoker
non−smoker

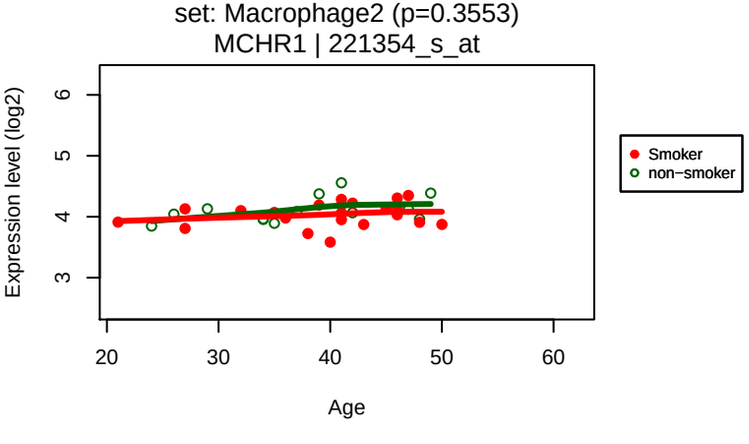

set: SmallAirway2 (p=0.1422)
MCHR1 | 221354_s_at
Expression level (log2)
Age
Smoker
non−smoker

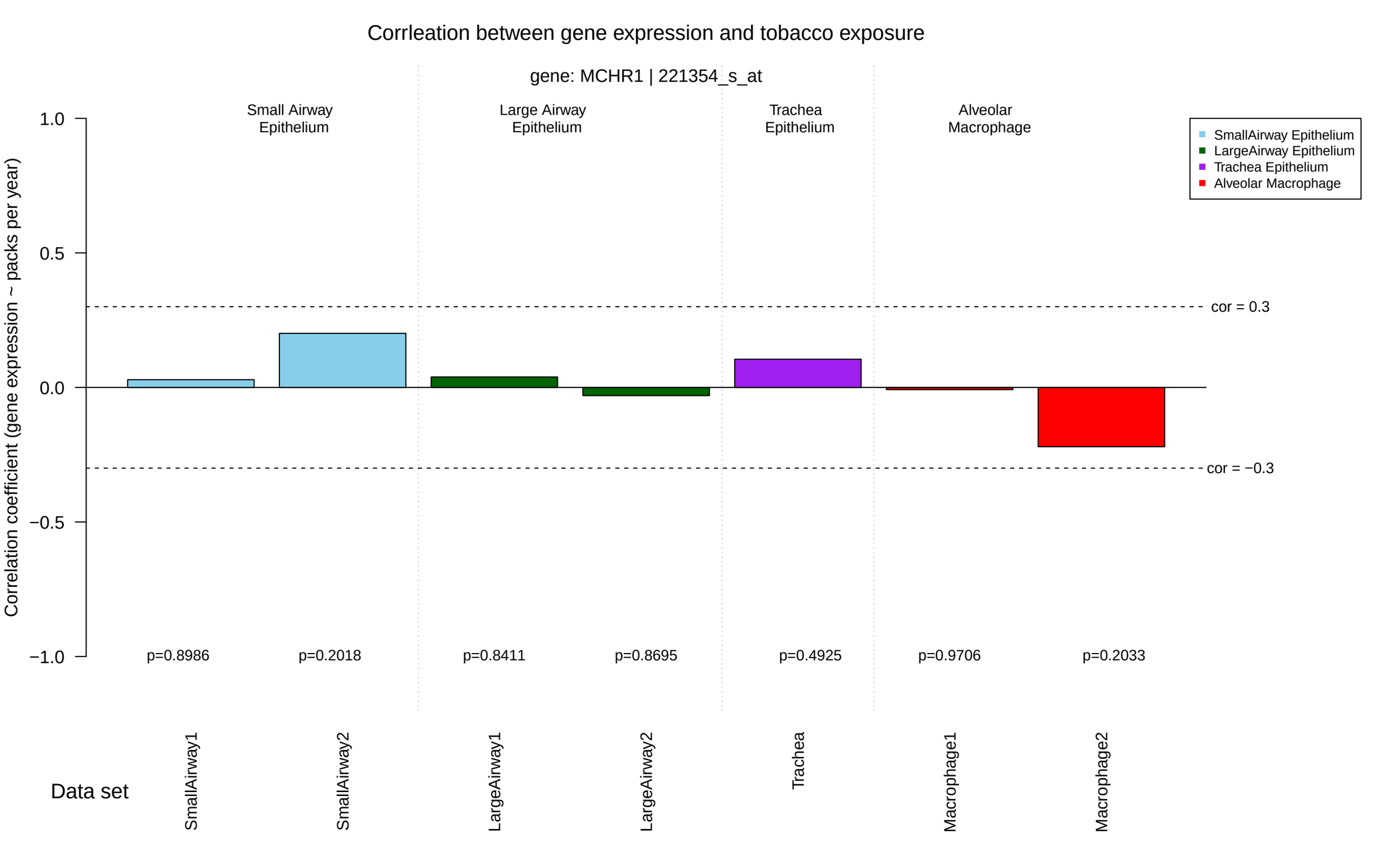

set: LargeAirway1 (p=0.3015)
MCHR1 | 221354_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6195)
MCHR1 | 221354_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.1088)
MCHR1 | 221354_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.7234)
MCHR1 | 221354_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3553)
MCHR1 | 221354_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: MCHR1 | 221354_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.8986
p=0.2018
p=0.8411
p=0.8695
p=0.4925
p=0.9706
p=0.2033
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2